Compact connected abelian groups with no small subgroups are tori

An informal account of Oliver Nash's Lean formalization

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Abstract

We explain Oliver Nash's Lean proof that a topological group is isomorphic to a torus if and only if it is compact, connected and abelian and has no small subgroups.

1 Statement

A topological group G has *no small subgroups* if some neighbourhood U of 1 contains no subgroup other than $\{1\}$.

Theorem 1. *A topological group G is isomorphic, as a topological group, to $\mathbb{T}^m = (S^1)^m$ for some m if and only if G is compact, connected and abelian and has no small subgroups.*

The easy direction. Since $\mathbb{T}^m$ is clearly compact, connected and abelian, it suffices to show that it has no small subgroups. Regard $\mathbb{T}^m$ as a subset of $\mathbb{C}^m$, and let $U = \{z \in \mathbb{T}^m : \operatorname{Re} z_j > 0 \text{ for all } j\}$. Let $H \subseteq U$ be a subgroup and $z \in H$, and suppose some coordinate $w = z_j$ is not 1. Write $w = e^{i\theta}$ with $0 < |\theta| \leq \pi$, and let k be the smallest positive integer with $k|\theta| \geq \pi/2$. Then $(k-1)|\theta| < \pi/2$, so $\pi/2 \leq k|\theta| < \pi/2 + |\theta| \leq 3\pi/2$ and $\operatorname{Re} w^k = \cos k\theta \leq 0$, contradicting $z^k \in U$. Hence $H = \{1\}$, and $\mathbb{T}^m$ has no small subgroups.

A *character* of a topological group G is a continuous homomorphism $G \to S^1$. Let $\widehat{G}$ denote the group of characters. For the hard direction, we use the following lemmas.

Lemma 2 (Characters separate points)**.** *If G is a compact Hausdorff abelian group and $g \neq 1$, there is a character χ with $\chi(g) \neq 1$.*

Lemma 3. *The characters of $\mathbb{T}^m$ are exactly the maps $z \mapsto \prod_{i=1}^m z_i^{n_i}$ with $n_1, \ldots, n_m \in \mathbb{Z}$. In particular, the characters of S^1 are exactly the maps $z \mapsto z^n$ with $n \in \mathbb{Z}$.*

Lemma 4. *A topological group G with no small subgroups is Hausdorff.*

2 Proof of the hard direction

Let G be compact, connected and abelian with no small subgroups, and fix an open neighbourhood U of 1 containing no subgroup other than $\{1\}$. By Lemma 4, G is Hausdorff. The set $K = G \setminus U$ is compact and misses 1. By Lemma 2, the open sets $\{\chi \neq 1\}$, $\chi \in \widehat{G}$, cover K, so finitely many $\chi_1, \ldots, \chi_k$ do.

If χ is a character with $\chi^n = 1$, then χ maps the connected space G into the n-th roots of unity, so $\chi = 1$. Thus $\widehat{G}$ is torsion-free, and the subgroup $M = \langle \chi_1, \dots, \chi_k \rangle$ is free abelian. Let $\psi_1, \dots, \psi_m$ be a basis of M, so that $L\colon \mathbb{Z}^m \to M$, $(n_1, \dots, n_m) \mapsto \prod_i \psi_i^{n_i}$, is an isomorphism, and let $\Psi = (\psi_1, \dots, \psi_m)\colon G \to \mathbb{T}^m$. We show that Ψ is an isomorphism.

Ψ is injective: By the choice of $\chi_1, \dots, \chi_k$,

$$K \subseteq \bigcup_{j=1}^{k} \{\chi_j \neq 1\} \implies \bigcap_{j=1}^{k} \ker \chi_j \subseteq G \setminus K = U \implies \bigcap_{j=1}^{k} \ker \chi_j = \{1\},$$

since U contains no nontrivial subgroup. Each χ_j is a product of powers of $\psi_1, \dots, \psi_m$, so $\chi_j(g) = 1$ whenever $\Psi(g) = 1$. Thus $\ker \Psi \subseteq \bigcap_j \ker \chi_j = \{1\}$.

Ψ is surjective: The image $H = \Psi(G)$ is compact, hence a closed subgroup of $\mathbb{T}^m$. Suppose, for the sake of contradiction, that $t \notin H$ for some $t \in \mathbb{T}^m$. Since H is closed, $\mathbb{T}^m/H$ is a compact Hausdorff abelian group, and $tH \neq 1$. By Lemma 2, there is a character ρ of $\mathbb{T}^m/H$ with $\rho(tH) \neq 1$, so $\tilde{\rho}(z) = \rho(zH)$ is a character of $\mathbb{T}^m$ with $\tilde{\rho}(H) = \{1\}$ and $\tilde{\rho}(t) \neq 1$. By Lemma 3, $\tilde{\rho}(z) = \prod_i z_i^{n_i}$ for some $n_i \in \mathbb{Z}$. Then

$$\begin{aligned} \tilde{\rho}(\Psi(G)) = \tilde{\rho}(H) = \{1\} &\implies \textstyle\prod_i \psi_i(g)^{n_i} = 1 \text{ for all } g \in G \\ &\implies \textstyle\prod_i \psi_i^{n_i} = 1 \text{ in } M \\ &\implies (n_1, \dots, n_m) = L^{-1}\textstyle(\prod_i \psi_i^{n_i}) = L^{-1}(1) = 0, \end{aligned}$$

so $\tilde{\rho}$ is trivial, contradicting $\tilde{\rho}(t) \neq 1$.

Thus Ψ is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism, and $G \cong \mathbb{T}^m$. □

3 Proof of Lemma 2

3.1 From Peter–Weyl

For a continuous unitary representation $\pi\colon G \to \mathrm{U}(\mathcal{H})$ on a finite-dimensional complex Hilbert space $\mathcal{H}$ and vectors $v, w \in \mathcal{H}$, the *matrix coefficient* $\pi_{v,w}\colon G \to \mathbb{C}$ is $\pi_{v,w}(x) = \langle \pi(x)v, w \rangle$. We use the Peter–Weyl theorem in the following form (see e.g. Folland, *A Course in Abstract Harmonic Analysis*, §5.2).

Theorem 5 (Peter–Weyl)**.** *If G is a compact Hausdorff group, the linear span of the matrix coefficients of its finite-dimensional continuous unitary representations is dense in $C(G)$ for the uniform norm.*

Proof of Lemma 2. Let $g \neq 1$.

A representation with $\pi(g) \neq 1$: Since G is compact Hausdorff, Urysohn's lemma gives $f \in C(G)$ with $f(1) = 1$ and $f(g) = 0$. By Theorem 5 there are finite-dimensional continuous unitary representations $\pi_1, \dots, \pi_r$ and vectors v_j, w_j such that $q = \sum_{j=1}^{r} (\pi_j)_{v_j, w_j}$ satisfies $\|f - q\|_\infty < 1/2$. Since $|f(1) - f(g)| = 1$, this forces $q(1) \neq q(g)$. If $\pi_j(g) = 1$ for every j, then $(\pi_j)_{v_j,w_j}(g) = \langle v_j, w_j \rangle = (\pi_j)_{v_j,w_j}(1)$ for every j, so $q(g) = q(1)$. Hence $\pi_j(g) \neq 1$ for some j. Put $\pi = \pi_j\colon G \to \mathrm{U}(\mathcal{H})$.

A character with $\chi(g) \neq 1$: Since G is abelian, the operators $\pi(x)$, $x \in G$, are commuting unitary operators on $\mathcal{H}$, so they are simultaneously diagonalisable. That is, $\mathcal{H}$ has an orthonormal basis $e_1, \dots, e_d$ with $\pi(x)e_i = \chi_i(x)e_i$ for all x and i. Each χ_i is multiplicative because π is a homomorphism, takes values in S^1 because $\pi(x)$ is unitary, and is continuous because $\chi_i(x) = \langle \pi(x)e_i, e_i \rangle$. So each χ_i is a character, and since $\pi(g) \neq 1$, some $\chi_i(g) \neq 1$. □

Peter–Weyl rests on the spectral theorem for compact self-adjoint operators. For abelian G the convolution operators of §3.2 commute, so the formalization can use Gelfand theory for commutative C*-algebras instead.

3.2 The formalized proof, via Gelfand theory

Setup. Let G be a compact Hausdorff abelian group and μ the unique regular Haar measure on G with $\mu(G) = 1$. The formalization allows any finite measure that is left- and inversion-invariant, weakly regular and positive on nonempty open sets; the normalisation only simplifies constants. For $f \in C(G)$, $h \in L^2(G, \mu)$ and $a \in G$ put

$$(f * h)(x) = \int_G f(xy^{-1})\, h(y)\, d\mu(y), \qquad f^\sharp(y) = \overline{f(y^{-1})}, \qquad (\tau_a f)(y) = f(ya^{-1}),$$

and let C_f be the convolution operator $h \mapsto f*h$ on L^2. Let $B(L^2)$ denote the C*-algebra of bounded linear operators on $L^2(G, \mu)$, with the operator norm and adjoint. Each $f \in C(G)$ is uniformly continuous, in the sense that for each $\varepsilon > 0$ there is a neighbourhood V of 1 with $|f(u) - f(v)| < \varepsilon$ whenever $uv^{-1} \in V$.

Lemma 6. *Let $f, f' \in C(G)$ and $a, b \in G$.*

(i) $C_f \in B(L^2)$, *and* $F \mapsto C_F$ *is a bounded linear map* $C(G) \to B(L^2)$.

(ii) $f * f' \in C(G)$, *and* $C_f C_{f'} = C_{f*f'} = C_{f'} C_f$.

(iii) $C_f^* = C_{f^\sharp}$.

(iv) $C_f \neq 0$ *whenever* $f \neq 0$.

(v) $\tau_a f * \tau_b f' = \tau_{ab}(f * f')$.

Let A_0 be the unital subalgebra of $B(L^2)$ generated by the C_f. By Lemma 6(ii) the C_f commute, so A_0 is commutative, and by Lemma 6(iii) it is closed under adjoints. Let A be the closure of A_0 in the operator norm. Since multiplication and adjoint are continuous, A is again a commutative unital $*$-subalgebra. Being closed in the complete space $B(L^2)$, it is complete, and it inherits the C*-identity $\|X^*X\| = \|X\|^2$ from $B(L^2)$. So A is a commutative C*-algebra. Let Δ be the set of nonzero multiplicative linear functionals $\varphi\colon A \to \mathbb{C}$, with the weak-$*$ topology, in which it is a compact Hausdorff space. By the commutative Gelfand–Naimark theorem, the Gelfand transform $A \to C(\Delta)$, $X \mapsto (\varphi \mapsto \varphi(X))$, is isometric, hence injective. So for $X \neq Y$ in A there is $\varphi \in \Delta$ with $\varphi(X) \neq \varphi(Y)$.

Proof of Lemma 2 via Gelfand theory. Let $g \neq 1$. By Urysohn choose a real $f \in C(G)$ with $f(g) = 0$ and $f(1) = 1$, and let $T(a) = C_{\tau_a f} \in A$. Then:

- T is continuous: Given $\varepsilon > 0$, choose V for f by uniform continuity. For $a, b \in G$ we have $(ya^{-1})(yb^{-1})^{-1} = a^{-1}b$, as G is abelian, so $\|\tau_a f - \tau_b f\|_\infty = \sup_y |f(ya^{-1}) - f(yb^{-1})| \leq \varepsilon$ whenever $b \in aV$. Thus $a \mapsto \tau_a f$ is continuous from G to $C(G)$ with the sup norm, and composing with the bounded linear map $F \mapsto C_F$ of Lemma 6(i) shows that T is continuous.
- $T(a)T(b) = T(ab)T(1)$: By Lemma 6(ii) and (v), both sides equal $C_{\tau_{ab}(f*f)}$.
- $T(g) \neq T(1)$: We have $T(g) - T(1) = C_{\tau_g f - f}$ and $(\tau_g f - f)(g) = 1$, so this is nonzero by Lemma 6(iv).

Pick $\varphi \in \Delta$ with $\varphi(T(g)) \neq \varphi(T(1))$ and set $c(a) = \varphi(T(a))$, so that $c(a)c(b) = c(ab)c(1)$. If $c(1) = 0$, then $c(a)^2 = c(a^2)c(1) = 0$ for all a, contradicting $c(g) \neq c(1)$. So $\chi = c/c(1)$ is a homomorphism $G \to \mathbb{C}^\times$, and it is continuous because T and φ are. Since G is compact, χ is bounded, say $|\chi| \leq R$. For $a \in G$ and $n \in \mathbb{Z}$ this gives $|\chi(a)|^n = |\chi(a^n)| \leq R$, which forces $|\chi(a)| = 1$. So χ takes values in S^1, it is a character, and $\chi(g) \neq 1$. □

A Proof of Lemma 3

Proof. Each map $z \mapsto \prod_j z_j^{n_j}$ is a continuous homomorphism, hence a character. We prove the converse, first for $m = 1$.

Let ψ be a character of S^1. Since $t \mapsto e^{it}$ is a covering map $\mathbb{R} \to S^1$ and $\psi(1) = 1$, the path lifting lemma gives a continuous $F\colon \mathbb{R} \to \mathbb{R}$ with $F(0) = 0$ and $\psi(e^{it}) = e^{iF(t)}$ for all t. Because ψ is a homomorphism,

$$e^{iF(s+t)} = \psi(e^{is}e^{it}) = \psi(e^{is})\,\psi(e^{it}) = e^{i(F(s)+F(t))},$$

so $F(s+t) = F(s) + F(t) + 2\pi k(s,t)$ with $k(s,t) \in \mathbb{Z}$. By continuity k is constant, and setting $s = t = 0$ gives $k = 0$. That is, F is additive. A continuous additive map is linear, so $F(t) = \lambda t$. Taking $t = 2\pi$ gives $e^{2\pi i\lambda} = \psi(1) = 1$, so $\lambda = n \in \mathbb{Z}$. Hence $\psi(e^{it}) = (e^{it})^n$ for all t, that is, $\psi(z) = z^n$.

Now let ψ be a character of $\mathbb{T}^m$, and let $\iota_j\colon S^1 \to \mathbb{T}^m$ be the inclusion in the j-th coordinate. Each $\psi \circ \iota_j$ is a character of S^1, so $\psi(\iota_j(w)) = w^{n_j}$ for some $n_j \in \mathbb{Z}$. Since $z = \iota_1(z_1) \cdots \iota_m(z_m)$ and ψ is a homomorphism,

$$\psi(z) = \prod_{j=1}^{m} \psi(\iota_j(z_j)) = \prod_{j=1}^{m} z_j^{n_j}.$$ □

B Proof of Lemma 4

Proof. Let U be a neighbourhood of 1 containing no subgroup other than $\{1\}$. Let y lie in the closure of $\{1\}$, so that every neighbourhood of y contains 1. Since left translation by y is a homeomorphism, yU^{-1} is a neighbourhood of y, so $1 \in yU^{-1}$, that is, $y \in U$. Thus the closure of $\{1\}$, which is a subgroup, lies in U. Since U contains no subgroup other than $\{1\}$, the closure is trivial, and $\{1\}$ is closed. Now let $x \neq x'$ in G. Then xx'^{-1} lies in the open set $G \setminus \{1\}$, so by continuity of $(a,b) \mapsto ab^{-1}$ there are neighbourhoods V of x and W of x' with $ab^{-1} \neq 1$ for all $a \in V$ and $b \in W$. Hence $V \cap W = \emptyset$, so G is Hausdorff. □

C Proof of Lemma 6

We use the notation of §3.2, and write $\int$ for $\int_G$ and dy for $d\mu(y)$, and similarly for other variables. Let $f, f' \in C(G)$ and $h \in L^2(G,\mu)$.

Proof. (F1) $\underline{f * h \text{ is continuous:}}$ For $x, x' \in G$, Cauchy–Schwarz and $\mu(G) = 1$ give

$$|(f*h)(x) - (f*h)(x')| \leq \sup_y |f(xy^{-1}) - f(x'y^{-1})|\,\|h\|_2.$$

Given $\varepsilon > 0$, choose V for f by uniform continuity. Taking $u = xy^{-1}$ and $v = x'y^{-1}$ in that condition gives $uv^{-1} = xx'^{-1}$, independent of y. So whenever $xx'^{-1} \in V$, we get $\sup_y |f(xy^{-1}) - f(x'y^{-1})| \leq \varepsilon$, and hence $|(f*h)(x) - (f*h)(x')| \leq \varepsilon\|h\|_2$. Since $\{x' : xx'^{-1} \in V\} = V^{-1}x$ is a neighbourhood of x and ε is arbitrary, $f * h$ is continuous at x.

(F2) $\underline{\|f*h\|_2 \le \|f*h\|_\infty \le \|f\|_\infty \|h\|_2}$: For $x \in G$, Cauchy–Schwarz applied to $y \mapsto f(xy^{-1})$ and h, together with invariance of μ, gives $|(f*h)(x)| \le \|f\|_2 \|h\|_2$. Since $\mu(G) = 1$, we have $\|\cdot\|_2 \le \|\cdot\|_\infty$ for bounded measurable functions, and applying this to $f*h$ and to f gives

$$\|f*h\|_2 \le \|f*h\|_\infty \le \|f\|_2 \|h\|_2 \le \|f\|_\infty \|h\|_2.$$

In particular $F \mapsto C_F$ is a bounded linear map $C(G) \to B(L^2)$.

(F3) $\underline{C_f C_{f'} = C_{f*f'}}$: Take h continuous. Then the integrand below is continuous on the compact space $G \times G$, hence a uniform limit of finite sums of products $\alpha(y)\,\beta(z)$ with $\alpha, \beta \in C(G)$ by Stone–Weierstrass, so the two iterated integrals agree. Thus

$$\begin{aligned}
(f*(f'*h))(x) &= \int f(xy^{-1}) \int f'(yz^{-1})\, h(z)\, dz\, dy \\
&= \int \Big(\int f(xy^{-1})\, f'(yz^{-1})\, dy \Big) h(z)\, dz && \text{(Fubini)} \\
&= \int \Big(\int f(xz^{-1}u^{-1})\, f'(u)\, du \Big) h(z)\, dz && (y = uz) \\
&= \int (f*f')(xz^{-1})\, h(z)\, dz = ((f*f')*h)(x),
\end{aligned}$$

where the substitution $y = uz$ preserves μ by translation invariance. By (F2) both sides are bounded in h, and $C(G)$ is dense in $L^2(G,\mu)$ because μ is regular, so this holds for all $h \in L^2$.

(F4) $\underline{f*f' = f'*f}$: The map $y \mapsto xy^{-1}$ is an inversion followed by a translation, so it preserves μ, and since G is abelian, $x(xy^{-1})^{-1} = y$. Substituting,

$$(f*f')(x) = \int f(xy^{-1})\, f'(y)\, dy = \int f(y)\, f'(xy^{-1})\, dy = (f'*f)(x).$$

(F5) $\underline{C_f^* = C_{f^\sharp}}$: For continuous h and k, Fubini (justified as in (F3)) and $\overline{f(xy^{-1})} = f^\sharp(yx^{-1})$ give

$$\langle f*h, k\rangle = \iint f(xy^{-1})\, h(y)\, \overline{k(x)}\, dy\, dx = \int h(y) \overline{\int f^\sharp(yx^{-1})\, k(x)\, dx}\, dy = \langle h, f^\sharp * k\rangle.$$

By (F2) both sides are continuous in (h,k), so by density this holds for all $h, k \in L^2$.

(F6) $\underline{C_f \neq 0 \text{ whenever } f \neq 0}$: Since μ gives positive mass to every nonempty open set, a continuous function that vanishes almost everywhere vanishes everywhere. Suppose $C_f = 0$. Then $f * f^\sharp = C_f f^\sharp$ vanishes almost everywhere, and it is continuous by (F1), so it vanishes everywhere. In particular

$$0 = (f*f^\sharp)(1) = \int f(y^{-1})\, \overline{f(y^{-1})}\, dy = \int |f(y^{-1})|^2\, dy,$$

so the continuous nonnegative function $y \mapsto |f(y^{-1})|^2$ vanishes almost everywhere, hence everywhere, and $f = 0$.

(F7) $\underline{\tau_a f * \tau_b f' = \tau_{ab}(f*f')}$: Substituting $y = ub$ and using that G is abelian,

$$(\tau_a f * \tau_b f')(x) = \int f(xy^{-1}a^{-1})\, f'(yb^{-1})\, dy = \int f(x(ab)^{-1}u^{-1})\, f'(u)\, du = (f*f')(x(ab)^{-1}).$$

Together these give Lemma 6. Part (i) is (F2), part (ii) follows from (F1), (F3) and (F4), and parts (iii), (iv) and (v) are (F5), (F6) and (F7). □